

1. Justify the following with reasoning, no marks for incorrect justification.

(?0)

[Reasoning/Justification: brevity is the key, it saves paper and time]

- (a) Your senior, who is now an engineer, asks your help in designing a system that performs spectrum analysis using DFTs. He/She states that the spectrum analysis system's input data sample rate, f_s , is 1000 Hz and specifies that the DFT's frequency-domain sample spacing must be exactly 45 Hz. What do you tell about your senior's (and also your) understanding regarding the spectrum analysis system's specifications, are the given specifications correct? Can you conclude that he/she was regular in lectures?
- (b) When using windowing and zero padding for a length- N sequence $x[n]$, which of the following two is the proper procedure: (1) zero-pad first to length M , then use a window of length M , or (2) use a length- N window on $x[n]$, then zero-pad to length M ? Give reasons.
- (c) Linear convolution between the channel input, $x[n]$, and impulse response, $h[n]$, converts to circular convolution by adding a CP in OFDM. Justify the above statement mathematically.
- (d) Assume we have a 72-sample sinusoidal $x[n]$ time-domain sequence, the first 36 samples of which are shown in Fig.(a). Next we decimate $x[n]$ by two to generate 36 samples of the $y[m]$ sequence shown in Fig.(b). Sequence $y[m]$ is also sinusoidal, as we should expect, but its frequency appears to be double the frequency of $x[n]$. Explain that apparent frequency difference.

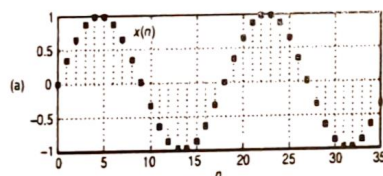


Fig. (a)

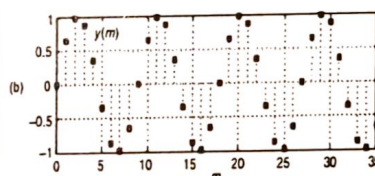
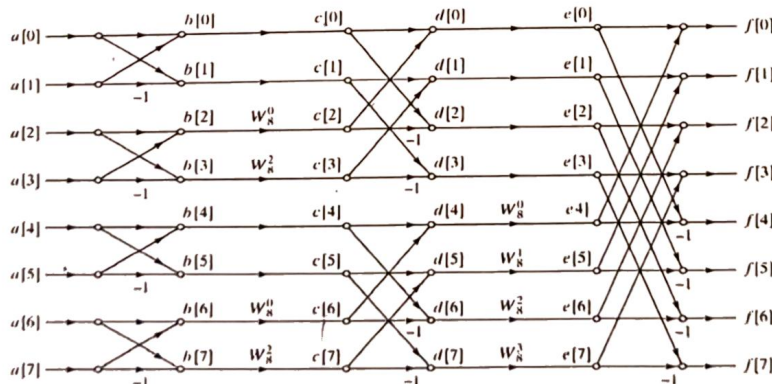


Fig. (b)

2. Consider the 8-point DIT FFT butterfly representation as shown, with a and f as input and output vectors, respectively. Assume that the input is in bit-reversed order and the output is in normal order (as in standard DIT FFT), and b, c, d , and e denote the intermediate vectors.
 - i) Determine the matrices F_1, T_1, F_2, T_2 , and F_3 such that $b = F_1 a, c = T_1 b, d = F_2 c, e = T_2 d, f = F_3 e$. The overall FFT can be described as $f = Q a$, with $Q = F_3 T_2 F_2 T_1 F_1$.
 - ii) Draw the butterfly structure for the sequence of operations described by Q^H , where $(\cdot)^H$ denotes complex (Hermitian) transpose of the matrix.
 - iii) What does this structure compute?

(?3)



3. Consider a 4th-order tapped-delay line FIR filter whose transfer function is $H[z] = b_0 + b_1 z^{-1} + b_2 z^{-2} + b_3 z^{-3} + b_4 z^{-4}$. Assume the filter has real-valued coefficients and that the filter is a linear-phase filter. If one of the filter's z -plane zeros has a value of $0.5657 + j0.5657$, what are the values of the other three z -plane zeros of this filter? (?¹)
4. For the given IIR transfer function, give the realization in the Gray-Markel form (?²)

$$H[z] = \frac{1 + 2z^{-1} + 3z^{-2}}{1 - z^{-1} + \frac{1}{4}z^{-2}}$$

5. You and your friend were given the following specifications for a digital filter

$$0.8 \leq |H(e^{j\omega})| \leq 1, \quad 0 \leq f \leq 800 \text{ Hz},$$

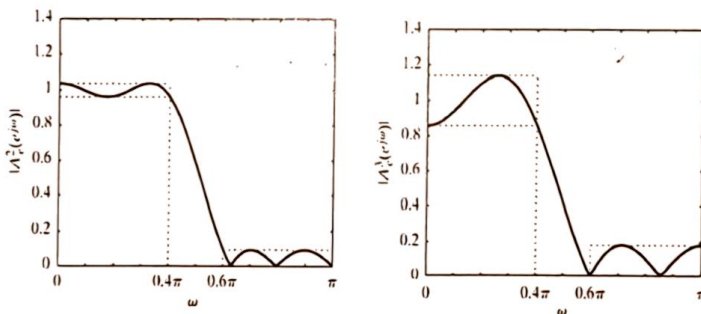
$$|H(e^{j\omega})| \leq 0.2, \quad 1040 \leq f \leq 4000 \text{ Hz}.$$

Using the BLT technique you computed the order of an IIR Chebyshev Type I digital filter to meet the given design specifications. Your friend calculated the order of an IIR Butterworth digital filter using BLT technique for the given design specifications. Comment on the order of the filter obtained by you and your friend and explain which design would you prefer: the one by you or the one by your friend. (?⁰)

6. For the system shown, find an expression for $y[n]$ in terms of $x[n]$. Simplify the expression as much as possible. (?⁴)



7. The graphs depict frequency-response magnitude plots of linear-phase FIR filters, labelled $|A_e^i(e^{j\omega})|$, $i = 2, 3$. What type(s) (I, II, III, IV) of linear-phase FIR filters can $|A_e^i(e^{j\omega})|$ correspond to, for $i = 2, 3$, list all possible choices, with proper reasoning. (?⁴)



For this part of the paper, marking scheme has not been defined. You are supposed to decipher and that gives you 8 marks (for solving the following bullet points and calculations showing total marks). Answer all these parts together with a heading as marking scheme.

- Compute the 4-point DFT of the length-8 signal $x[n] = [1, 2, -2, 3, 4, -2, -1, 1]^T$ by computing the DFT of its mod-4 reduction. Then, real part of the second term of the DFT obtained gives $(?^0)$ and sum of first and third term of the DFT obtained gives $(?^4)$.
- Find the circular convolution of sequences $x_1[n] = [-3, 2, 4]$ and $x_2[n] = [2, -4, 0, 1]$. Then, the absolute value of the sum of the terms obtained gives $(?^1)$ and half of the second term gives $(?^3)$. Further, half of $(?^3)$ gives $(?^2)$.

Birla Institute of Technology and Science, Pilani
Digital Signal Processing (ECE/EEE F434)

Comprehensive Examination, Open Book, First Semester 2025 – 26

Maximum Time: 60 minutes

05/12/25

Maximum Marks: 30

1. Let $x[n]$ be a discrete-time signal on $0 \leq n \leq N - 1$ and $X^d[k]$ its DFT.

(a) Define

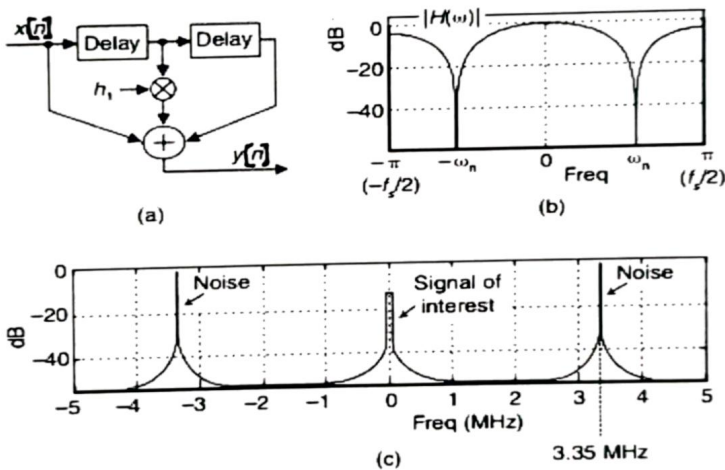
$$Y_1^d[k] = 0.42X^d[k] - 0.25X^d[(k-1) \bmod N] - 0.25X^d[(k+1) \bmod N] \\ + 0.04X^d[(k-2) \bmod N] + 0.04X^d[(k+2) \bmod N].$$

Show that $Y_1^d[k]$ represents a windowing operation in the time domain. To which of the windows you have learned is this window related? (4)

(b) Discuss possible implementation advantages of the window in part (a). (2)

(c) Can the idea proposed in this problem be applied to other windows you know? (2)

2. A 3-tap nonrecursive FIR filter is shown in Fig.(a). If $|h_1| \leq 2$, the filter will have an $|H(\omega)|$ frequency magnitude response having two nulls at $\pm\omega_n$ as shown in Fig.(b). (Here, the frequency axis value of π radians/sample corresponds to a cyclic frequency of half the sample rate, $f_s/2$.)



- (a) Assume we have a low-frequency signal of interest that's contaminated with high-level narrowband noise located at ± 3.35 MHz when the sample rate is $f_s = 8.25$ MHz as shown in Fig.(c). To attenuate that noise, for what value of h_1 , will the 3-tap FIR filter's nulls be located at the noise center frequency of ± 3.35 MHz? Show all the steps. (6)

(b) What is the DC gain of our 3-tap FIR filter? (2)

(c) Explain why the filter has a linear, or nonlinear, phase response. (2)

3. Let $x[n]$ be an N -point sequence such that $x[n] = 0$ for $n < 0$ and for $n > N - 1$. Let $\hat{x}[n]$ be the $2N$ -point sequence obtained by repeating $x[n]$, i.e.,

$$\hat{x}[n] = \begin{cases} x[n], & 0 \leq n \leq N - 1, \\ x[n - N], & N \leq n \leq 2N - 1. \end{cases}$$

Consider implementation of DT filter as shown in Figure 1 and Figure 2 on the next page. This system has an impulse response which is given as $h[n] = 0$ for $n < 0$ and $n > 2N - 1$.

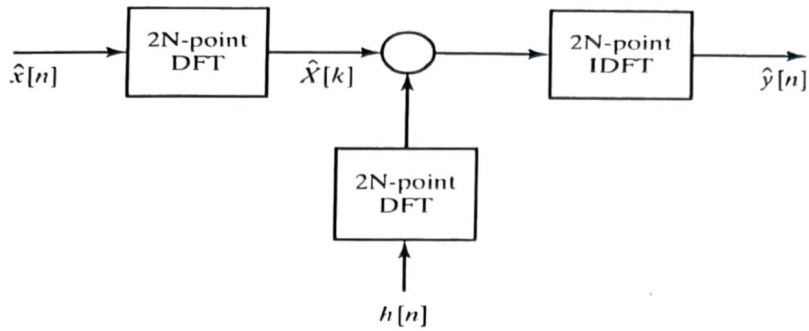


Figure 1: DT filter.

- (a) In Figure 1, what is $\hat{X}[k]$ in terms of $X[k]$. (6)
- (b) The system shown in Figure 1 can be implemented using only N -point DFTs as shown in Figure 2 for appropriate choices of System A and System B . Specify System A and System B so that $\hat{y}[n]$ in Figure 1 and $y[n]$ in Figure 2 are equal for $0 \leq n \leq 2N - 1$. (6)

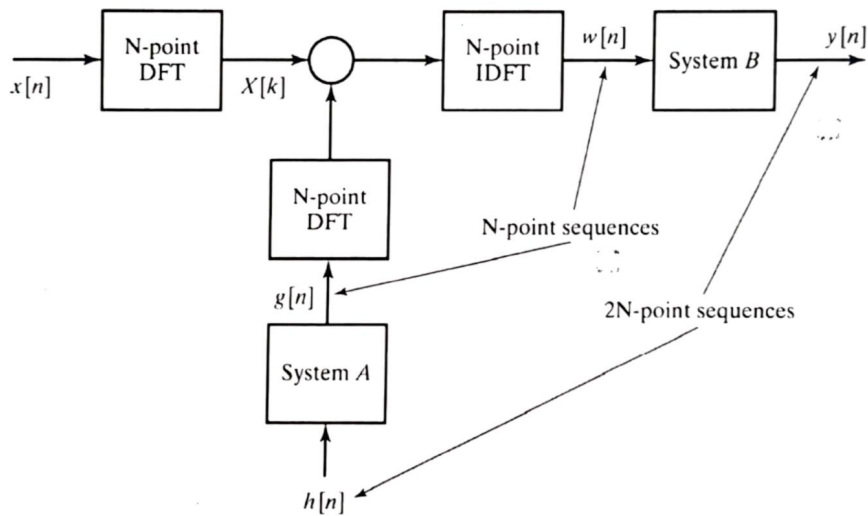


Figure 2: DT filter.